Second quantization, mean-field and spontaneous symmetry breaking

Today’s learning outcomes

- Describe what is a quantum material
- Explain why do we need quantum mechanics to understand quantum materials
- Understand the basics of second quantization
- Rationalize the basics of mean-field and spontaneous symmetry breaking

Control of materials shape technological ages

Stone age



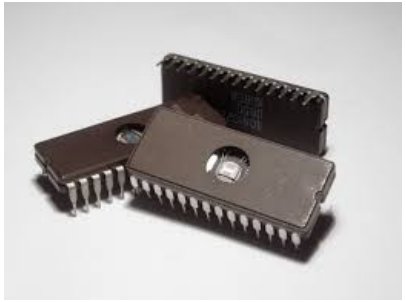
8700 BCE

Copper age



3500 BCE

Silicon age



20th century

Quantum age



21th century

Some of the future materials and technologies will rely on controlling quantum properties of matter

The impact of quantum materials

For medicine



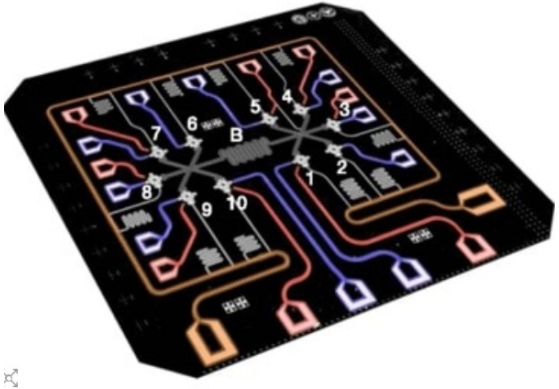
Superconductors

For renewable energies



Semiconductors

For quantum computing



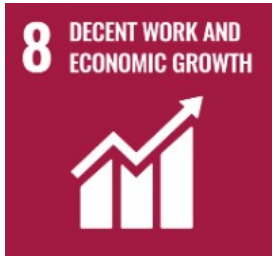
Semiconductors & superconductors

The UN sustainable development goals



Quantum materials and UN sustainable development goals





Quantum computing creates a whole new industry based on quantum materials





Quantum materials provide the platform for solar energy and future low consumption electronics



The development of quantum materials requires wide international cooperation

Quantum materials and UN sustainable development goals

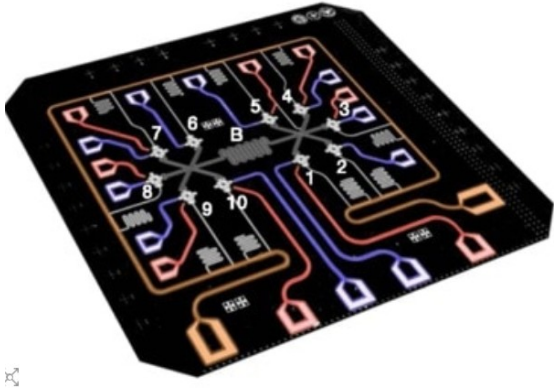


For renewable energies



Semiconductors

For quantum computing

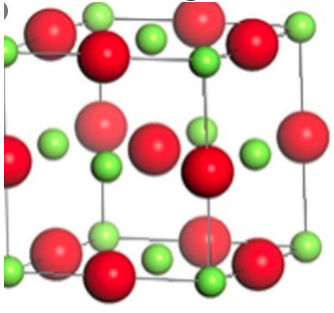


Semiconductors & superconductors

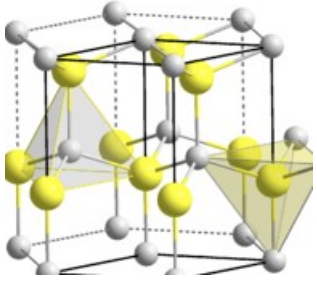


Which materials are we aiming to understand?

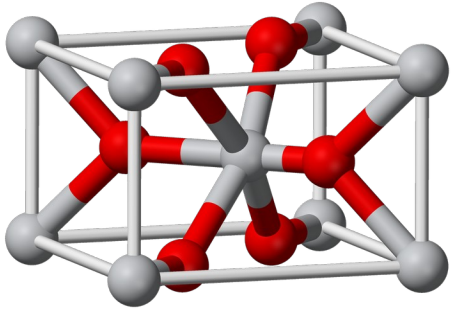
Insulators
MgO



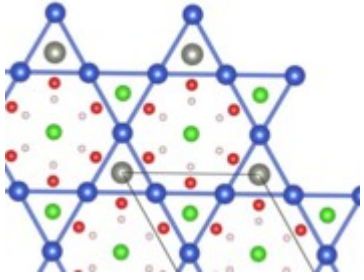
Semiconductors
GaN



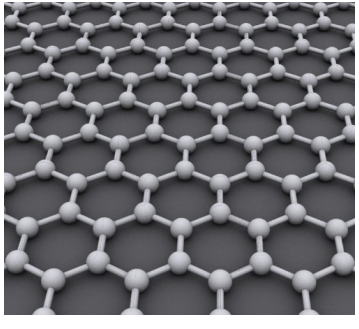
Ferromagnets
CrO_2



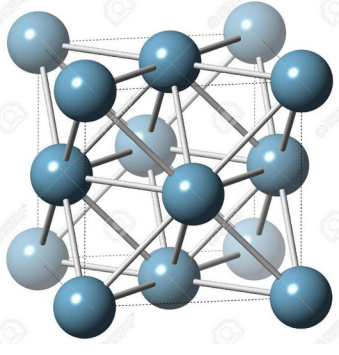
Spin liquids
Herbertsmithite



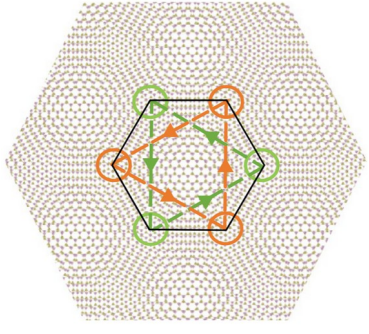
Dirac matter
graphene



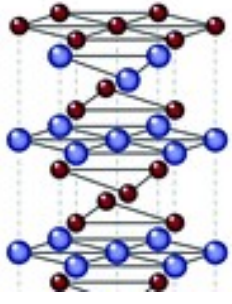
Superconductors
Al



Fractional matter
WTe_2



Topological insulators
Bi_2Se_3

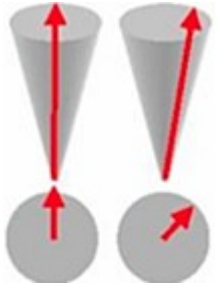


Quantum excitations in quantum materials

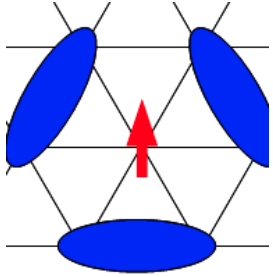
Solid state matter is made of electrons, protons, neutrons and photons

But in solid state materials, we can have emergent collective new excitations

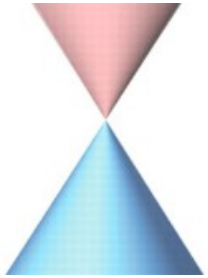
Magnons



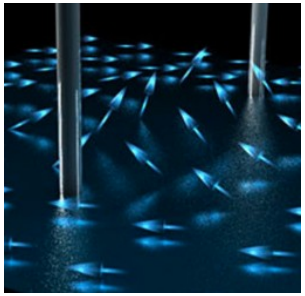
Spinons



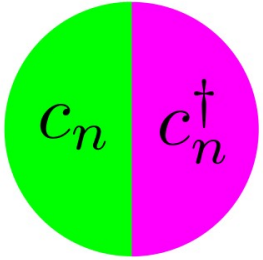
Dirac fermions



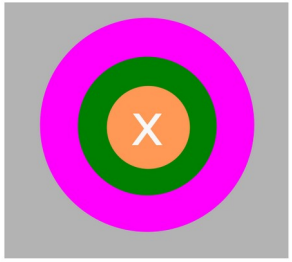
Vortexes



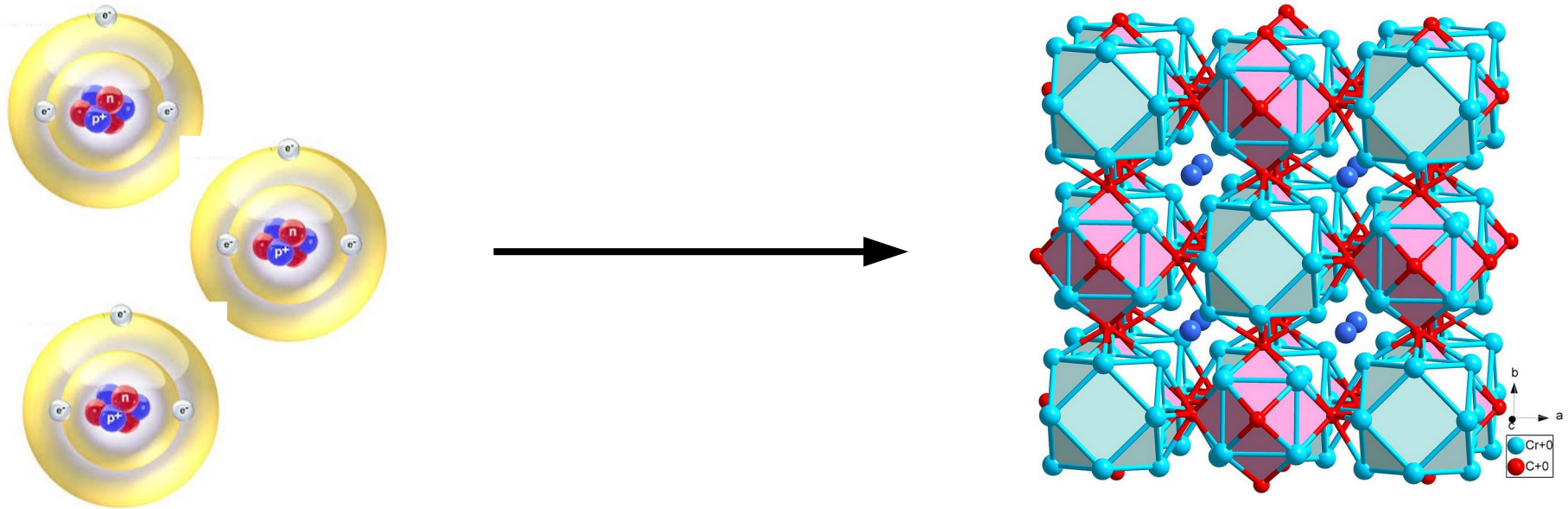
Majorana fermions



Anyons



From atoms to quantum materials



How do we understand and predict properties as we put more and more atoms together?

How do we describe quantum matter?

We use quantum mechanics to understand electrons in materials

$$H|\Psi\rangle = -i\partial_t|\Psi\rangle$$

Hamiltonian

Wavefunction of the system

Two main kinds of phenomena can emerge

Single particle phenomena

Many-body phenomena

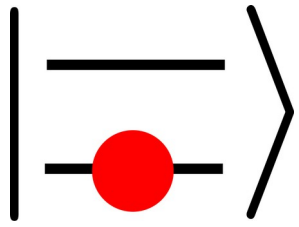
Two different kinds of quantum mechanical formalism

Systems where our number of particles is constant

First quantization, description based on a Hilbert space

Describes metals, semiconductors, insulators

Highly successful and easy formalism

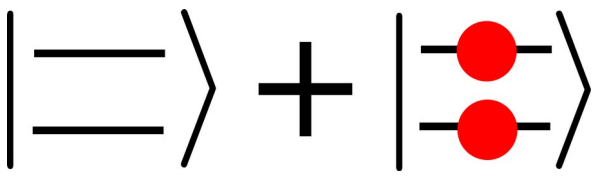


Systems where the number of particles fluctuates

Second quantization, description based on a Fock space

Describes superconductors, superfluids, correlated matter

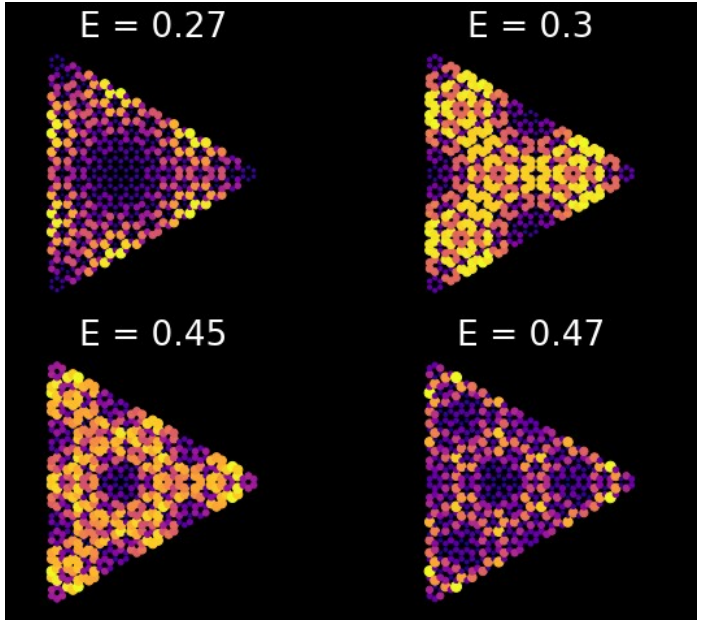
Leads to much exotic phenomena, yet also more challenging

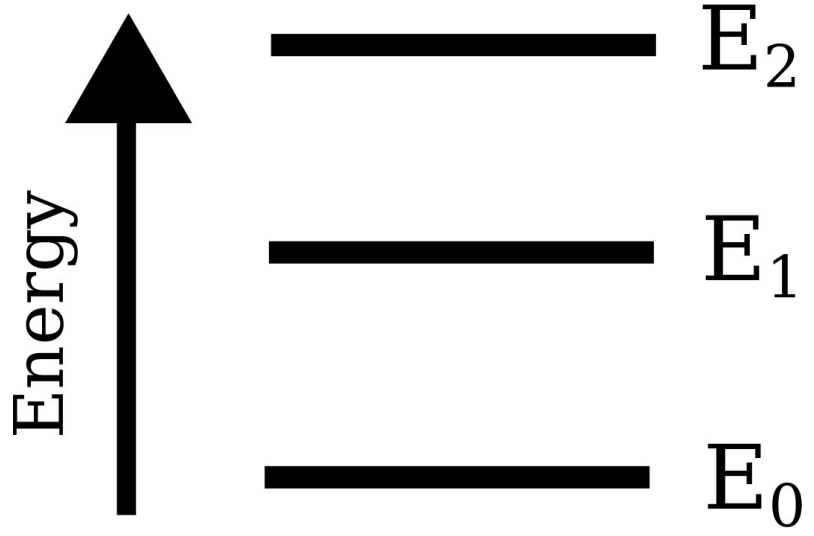


A reminder of a simple single particle state

Particle in a box

$$H|\Psi\rangle = E_n|\Psi\rangle$$

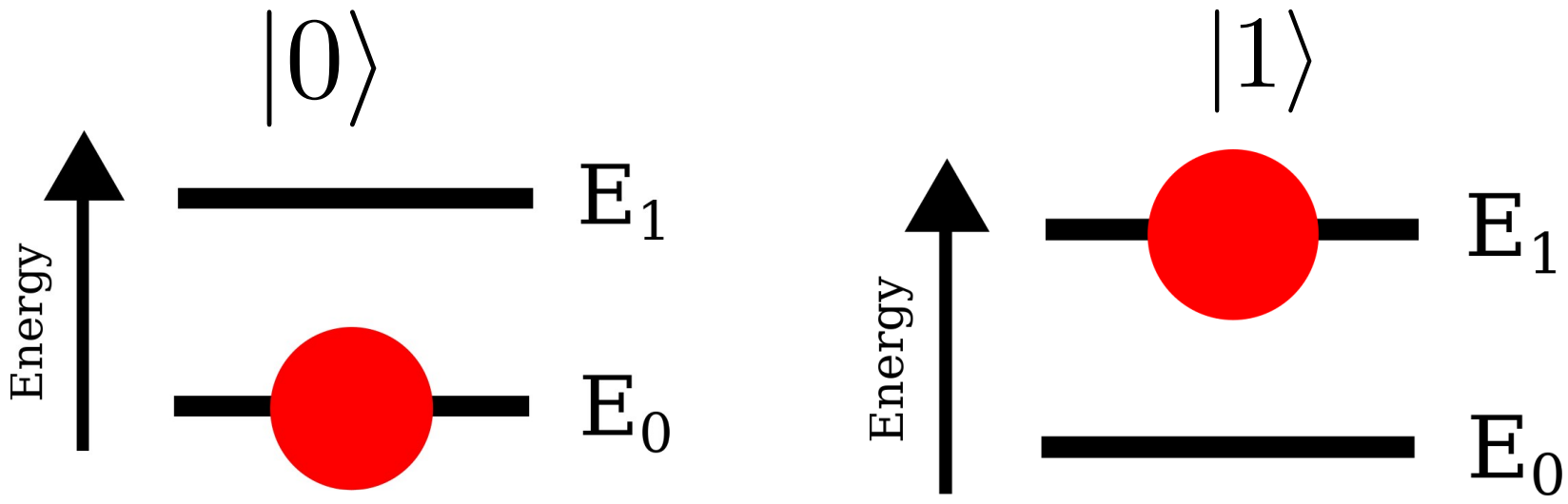




A reminder of a simple single particle state

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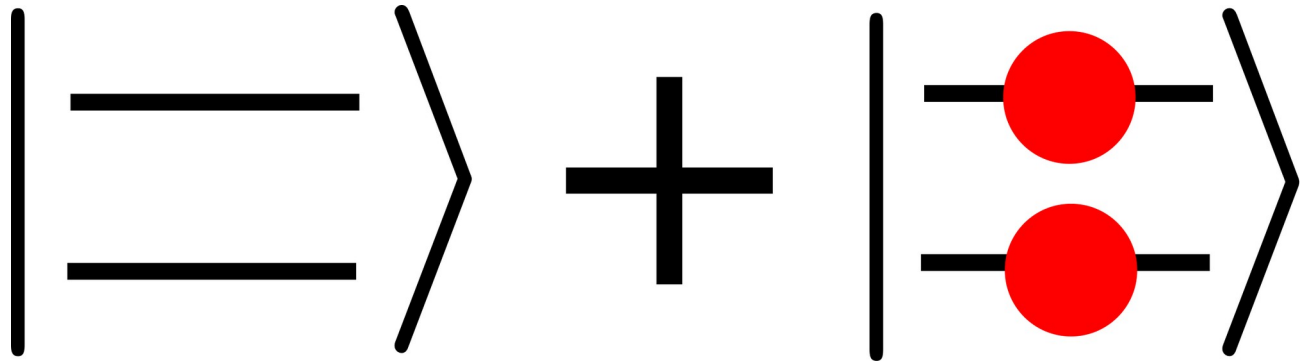


These two states describe having one particle, in one of the possible energy level

From single particle to many body

But what if our state is a combination of states with different numbers of particles?

A state having both 0 particles and 2 particles

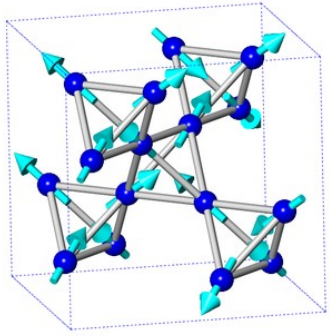


How do we describe states like these?

Non-constant number of particles

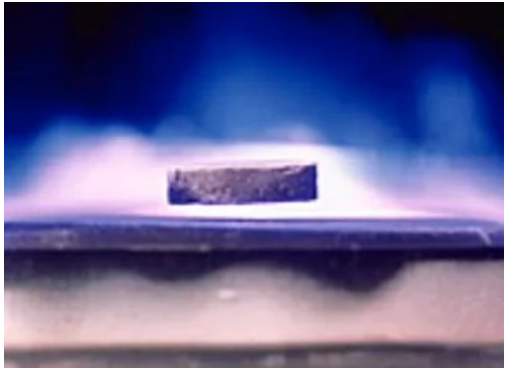
Why is this even relevant to understand materials?

Magnetic ordering of materials depends on processes in which the number of electrons fluctuates



Goodenough-Kanamori rules

The effective description of superconductors does not conserve electron number



Bogoliubov-de Gennes formalism

The idea of second quantization

Define operators that can create or destroy particles

c_i Annihilation operator, destroys a particle in site i

$c_i^\dagger$ Creation operator, creates a particle in site i

The empty vacuum state $|\Omega\rangle$ is defined as $c_i|\Omega\rangle = 0$

The Hamiltonian is written in terms of creation and annihilation operators

$$H = c_0^\dagger c_1 + h.c.$$

Fermionic quantum statistics in second quantization

Fermi-Dirac statistics for electrons

→ Wavefunctions are antisymmetric with respect to interchanging labels

→ There can only be 0 or 1 fermion per level

$$\{c_i^\dagger, c_j\} = c_i^\dagger c_j + c_j c_i^\dagger = \delta_{ij} \qquad \{c_i, c_j\} = 0$$

Anti-symmetric wavefunction $c_0^\dagger c_1^\dagger |\Omega\rangle = -c_1^\dagger c_0^\dagger |\Omega\rangle$

At most one fermion per site $c_0^\dagger c_0^\dagger |\Omega\rangle = 0$

The idea of second quantization

Lets see some examples using the two-levels presented before

$$|\Omega\rangle = \left|\begin{array}{c}\text{———}\\ \text{———}\end{array}\right\rangle$$ The “vacuum” state

$$|0\rangle \equiv c_0^\dagger|\Omega\rangle = \left|\begin{array}{c}\text{———}\\ \text{●——}\end{array}\right\rangle$$ One particle in level #0

$$|1\rangle \equiv c_1^\dagger|\Omega\rangle = \left|\begin{array}{c}\text{●——}\\ \text{———}\end{array}\right\rangle$$ One particle in level #1

$$c_0^\dagger c_1^\dagger|\Omega\rangle = \left|\begin{array}{c}\text{●——}\\ \text{●——}\end{array}\right\rangle$$ Two particles in level #0 & #1

First and second quantization

First quantization

$$H = \sum_{ij} t_{ij} |i\rangle\langle j|$$

$$H = \sum_{\alpha} \epsilon_{\alpha} |\Psi_{\alpha}\rangle\langle\Psi_{\alpha}|$$

$$|\Psi_{\alpha}\rangle = \sum_{n} U_{\alpha,n} |n\rangle$$

Second quantization

$$H = \sum_{ij} t_{ij} c_i^{\dagger} c_j$$

$$H = \sum_{\alpha} \epsilon_{\alpha} \Psi_{\alpha}^{\dagger} \Psi_{\alpha}$$

$$\Psi_{\alpha}^{\dagger} = \sum_{n} U_{\alpha,n} c_n^{\dagger}$$

Different kinds of Hamiltonians

Single particle Hamiltonians $H = \sum_{ij} t_{ij} c_i^\dagger c_j$

Insulators, semiconductors, metals

Many-body Hamiltonian $H = \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$

Fractional quantum Hall states, superconductors, quantum magnets

With second quantization, both cases can be treated on the same footing

Solving single particle Hamiltonians

A generic single particle Hamiltonian takes the form

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j$$

We can define a unitary transformation in the field operators as $\psi_\alpha^\dagger = \sum_n U_{\alpha,n} c_n^\dagger$

That diagonalizes the Hamiltonian $H = \sum_\alpha \epsilon_\alpha \psi_\alpha^\dagger \psi_\alpha \qquad UU^\dagger = I$

The single particle many-body ground state becomes $|GS\rangle = \prod_{\epsilon_\alpha < 0} \psi_\alpha^\dagger |\Omega\rangle$

With ground state energy $E_{GS} = \langle GS|H|GS\rangle = \sum_{\epsilon_\alpha < 0} \epsilon_\alpha$

Characterizing single particle Hamiltonians

Let us take a Hamiltonian in its diagonal form

$$H = \sum_\alpha \epsilon_\alpha \psi_\alpha^\dagger \psi_\alpha$$

$$\psi_\alpha^\dagger = \sum_n U_{\alpha,n} c_n^\dagger$$

There are as many single particle eigenvalues as sites (which can be a very large number)

We can characterize the single particle spectra by the energy distribution of eigenvalues

Density of states $$D(\omega) = \sum_\alpha \delta(\omega - \epsilon_\alpha)$$

And the spatial distribution of eigenstates by their spatially resolved energy distribution

Local density of states $$\rho(\omega, n) = \sum_\alpha \delta(\omega - \epsilon_\alpha) |\langle \psi_\alpha | n \rangle|^2$$

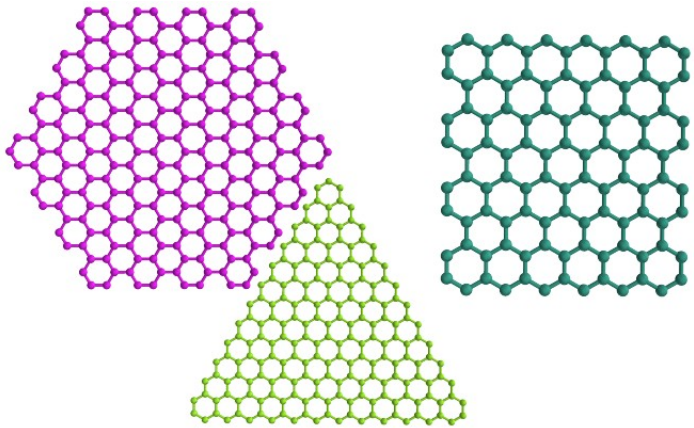
Modes in a single particle Hamiltonian

To solve a single particle Hamiltonian, we just have to diagonalize the matrix

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j$$

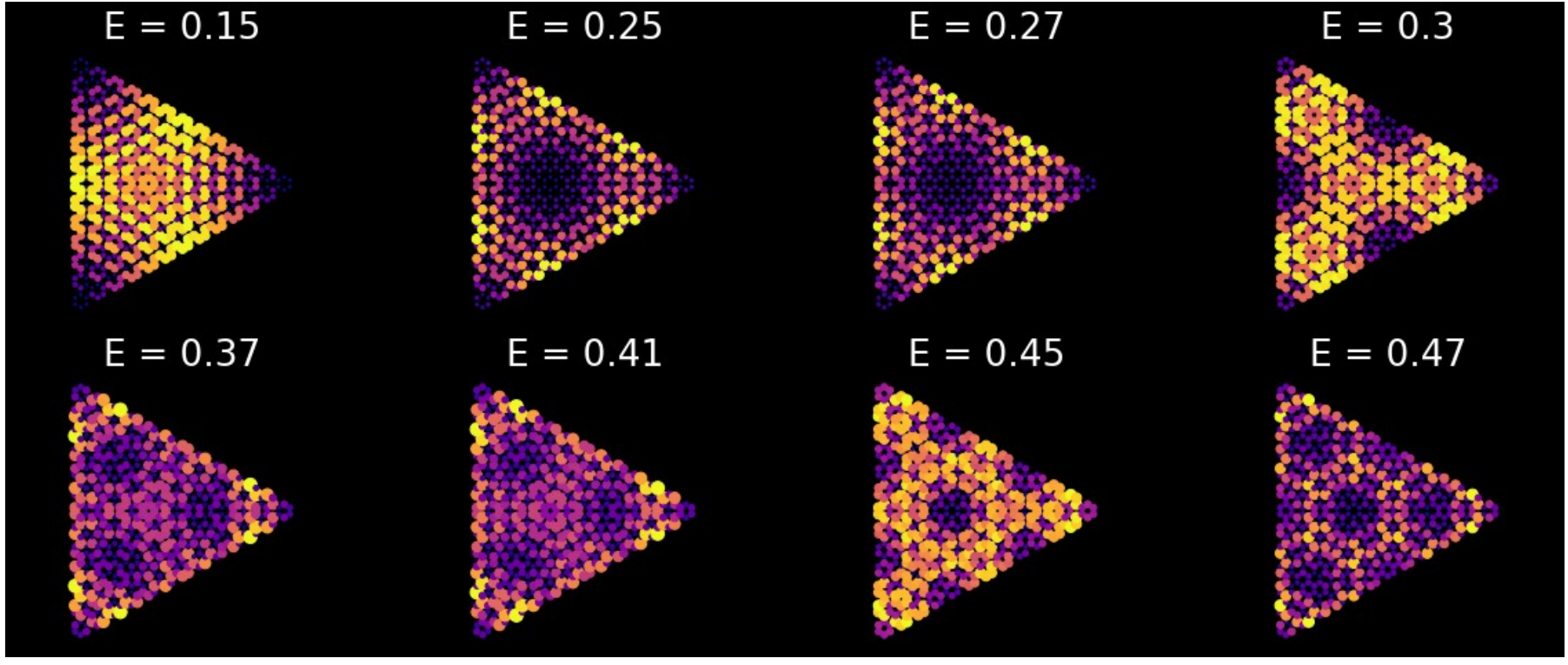
$$H = \sum_{\alpha} \epsilon_\alpha \psi_\alpha^\dagger \psi_\alpha$$

If take a certain finite geometry, we can find the confined modes

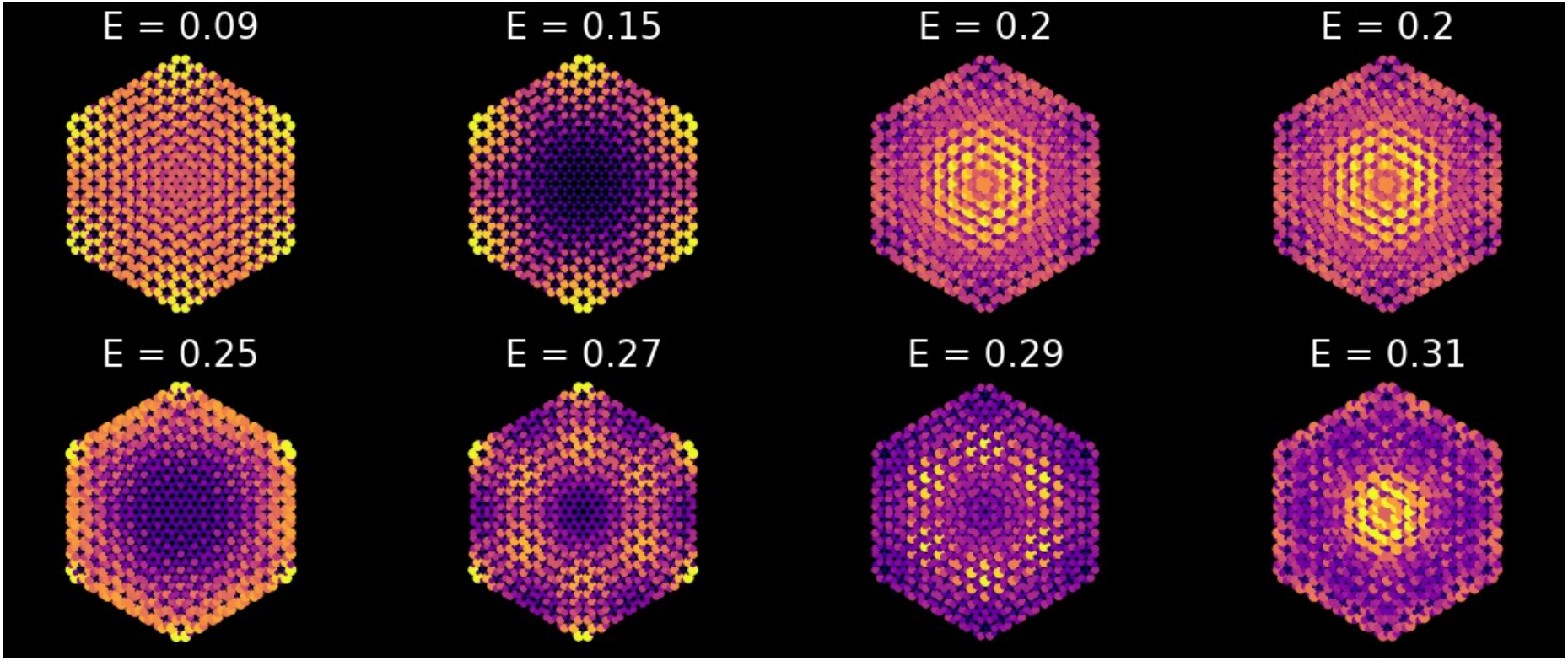


Let us do it for graphene islands

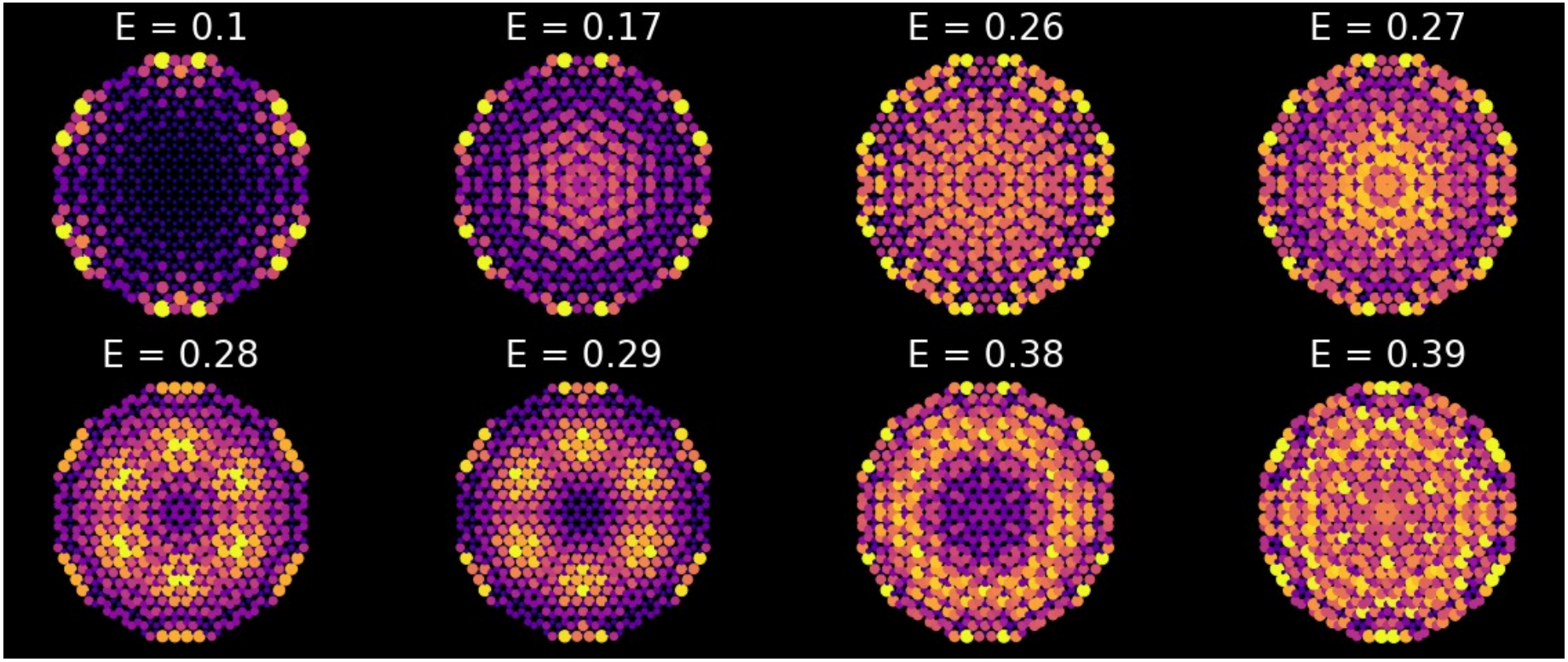
Confined modes in graphene islands



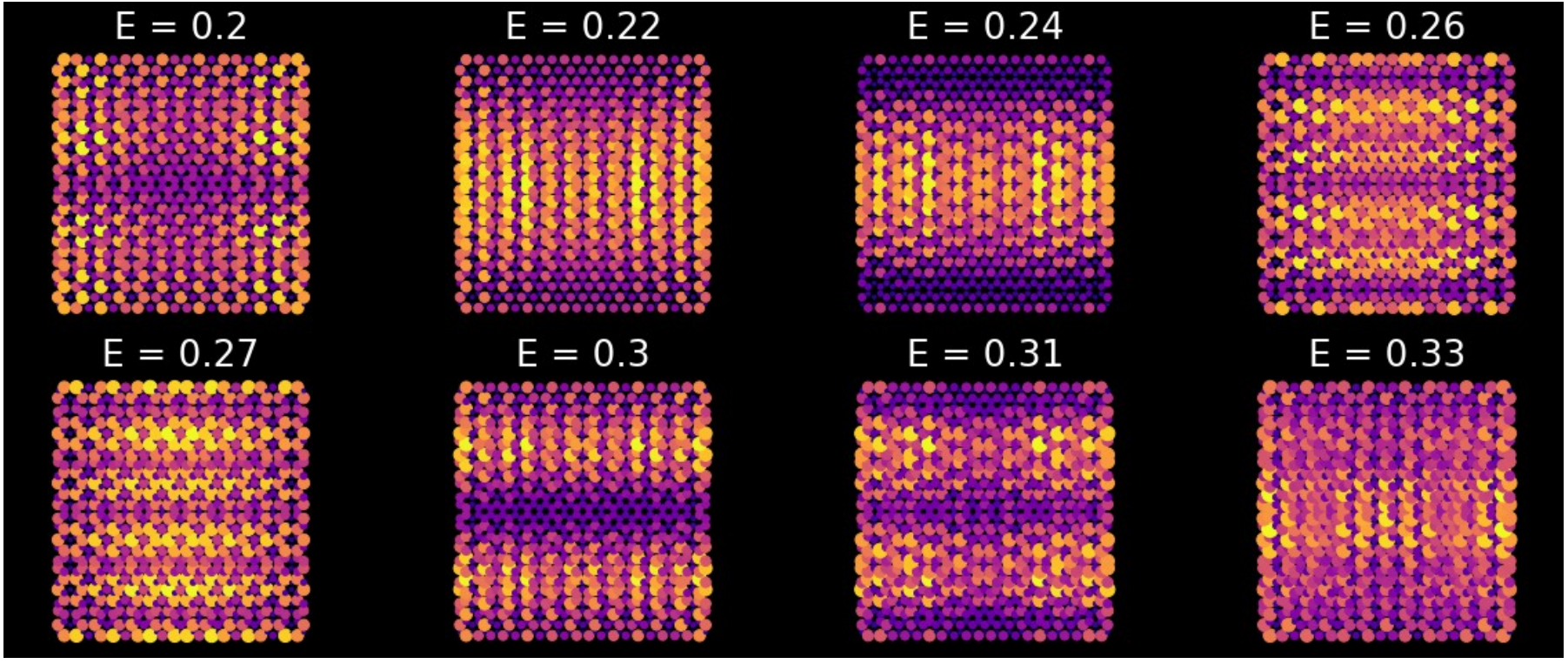
Confined modes in graphene islands



Confined modes in graphene islands



Confined modes in graphene islands



Solving tight binding models

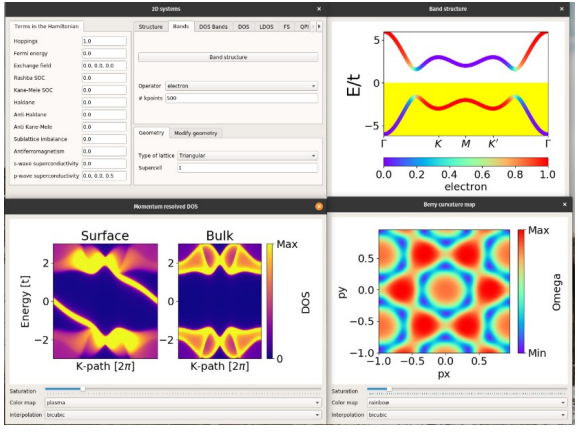
pyqula

```
from pyqula import geometry
g = geometry.honeycomb_lattice()
h = g.get_hamiltonian()
h.add_rashba(0.2) # Rashba spin-orbit coupling
h.add_zeeman([0.,0.,0.6]) # Zeeman field
from pyqula import topology
(kx,ky,omega) = h.get_berry_curvature() # compute Berry curvature
c = h.get_chern() # compute the Chern number
```

- Python library
- Ideal for complex models/calculations
- For writing in Python

https://github.com/joselado/pyqula

Quantum-lattice



- User-friendly interface for tight binding models
- Ideal for simple models and quick checks
- Fully interface-based, no scripting

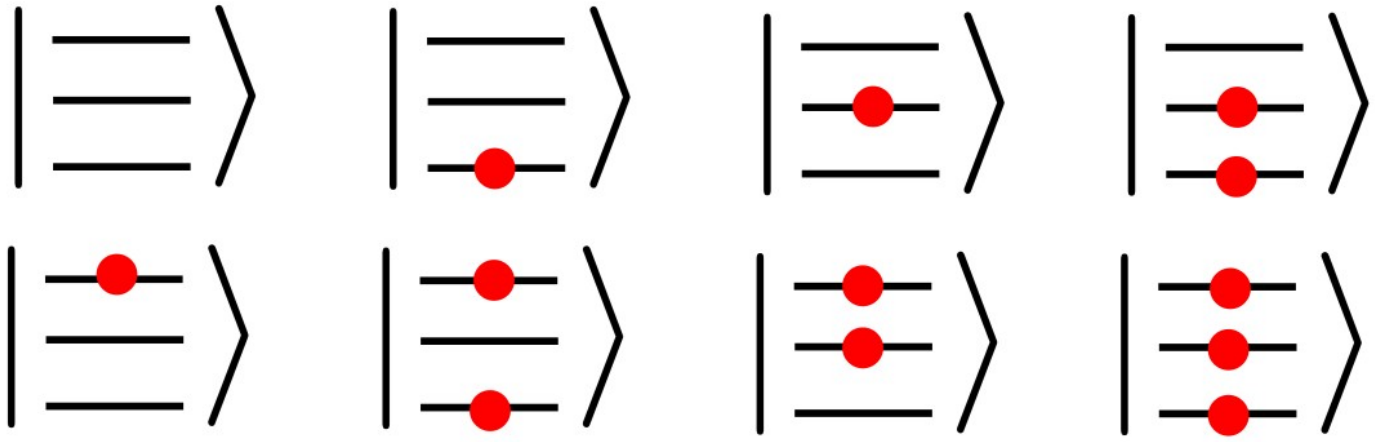
https://github.com/joselado/quantum-lattice

The fermionic quantum many-body problem

Let us take a minimal interacting fermionic Hamiltonian with 3 sites

$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + V \sum_n c_n^\dagger c_n c_{n+1}^\dagger c_{n+1}$$

The different many-body states are



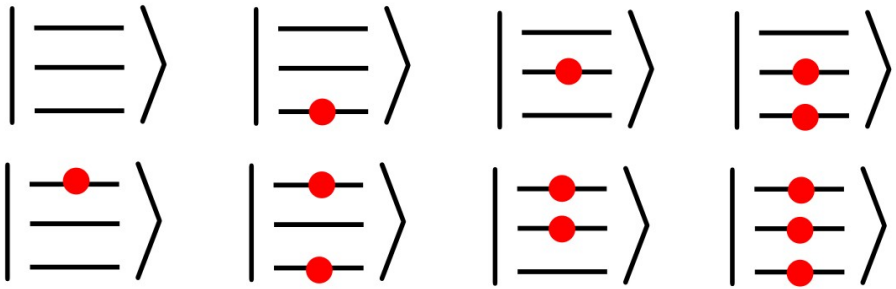
The fermionic quantum many-body problem

For a fermionic many-body problem with L sites

$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + V \sum_n c_n^\dagger c_n c_{n+1}^\dagger c_{n+1}$$

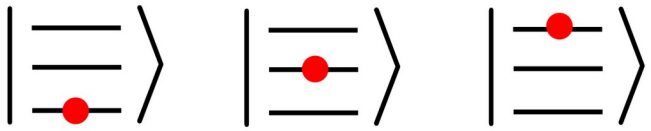
The dimension of the full Hilbert space grows as

$$d = 2^L$$



The dimension of the single particle space grows as

$$d = L$$



The fermionic quantum many-body problem

For an interacting fermionic many-body problem

$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + V \sum_n c_n^\dagger c_n c_{n+1}^\dagger c_{n+1}$$

A typical wavefunction is written as

$$|\Psi\rangle = \sum c_{n_1,n_2,...,n_L} |n_1, n_2, ...n_L\rangle$$

We need to determine in total 2^L coefficients

The Hamiltonian is a $2^L \times 2^L$ matrix

Three types of many-body sites

Spinful fermions



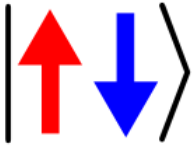
fully empty



down electron



up electron

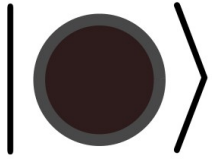


fully filled

Spinless fermions

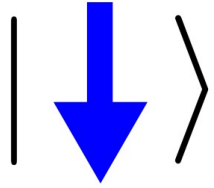


empty

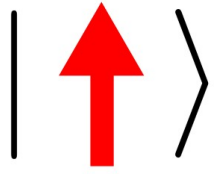


filled

Spins



down



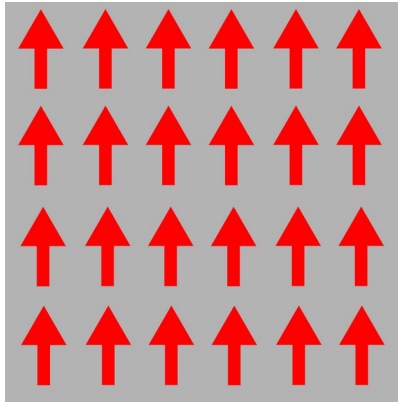
up

Many-body interactions and emergent states

Some quantum states can be approximately described by mean field theories

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

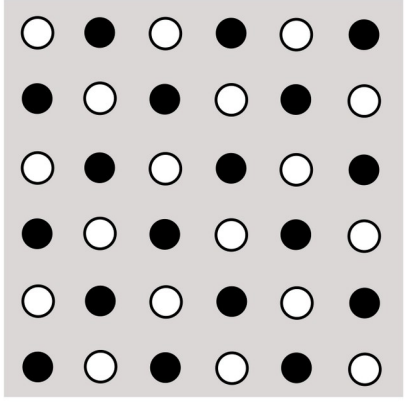
Magnets



Hubbard interaction

$$H \sim c_{\uparrow,n}^\dagger c_{\uparrow,n} c_{\downarrow,n}^\dagger c_{\downarrow,n}$$

Charge density waves



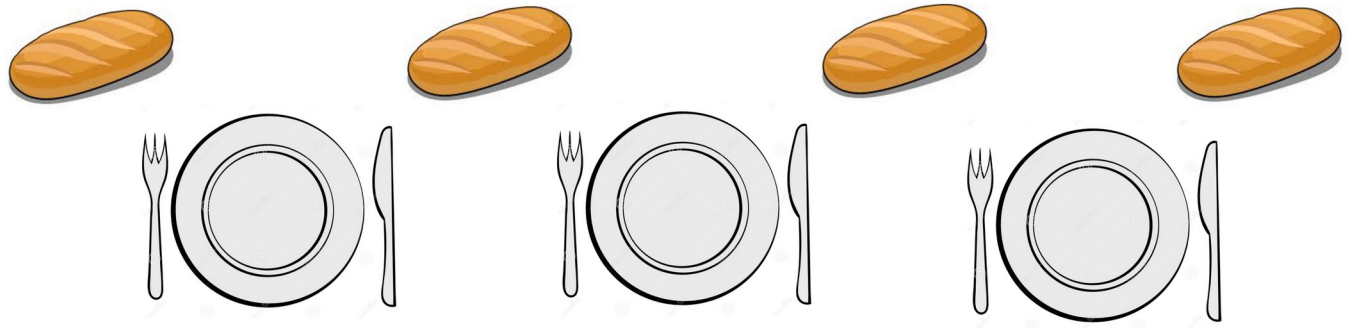
Nearest-neighbor repulsion

$$H \sim c_n^\dagger c_n c_{n+1}^\dagger c_{n+1}$$

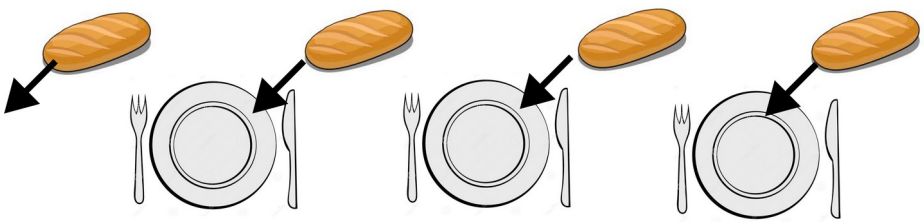
Spontaneous symmetry breaking

An example of symmetry breaking

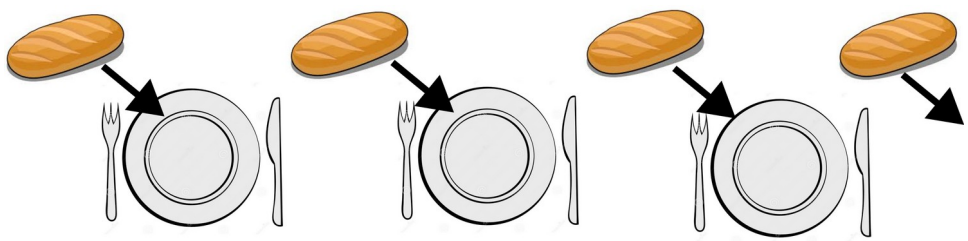
Pick your bread in a group meal



Solution #1



Solution #2



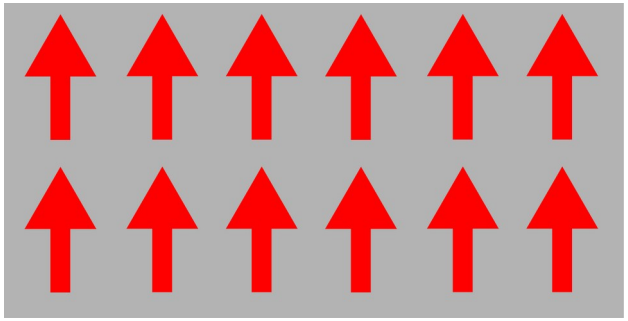
Spontaneous symmetry breaking

Central idea: a ground state can break the symmetry of a Hamiltonian

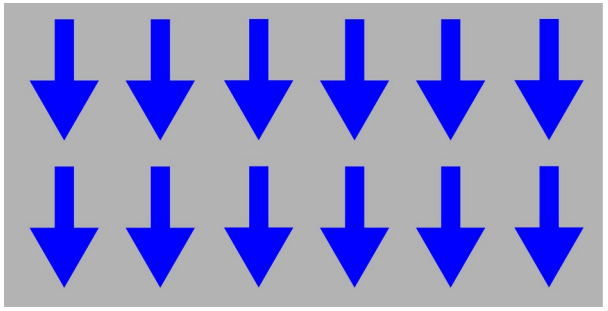
It can happen in the thermodynamic limit (infinite particles)

Hamiltonian for a ferromagnet $H = -\sum_{ij} \vec{S}_i \cdot \vec{S}_j$

Solution #1



Solution #2

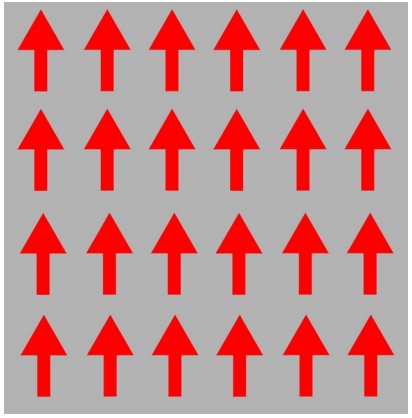


Nature (fluctuations) will choose one of the ground states as the macroscopic one

Classifying quantum matter according to symmetries

Broken
time-reversal symmetry

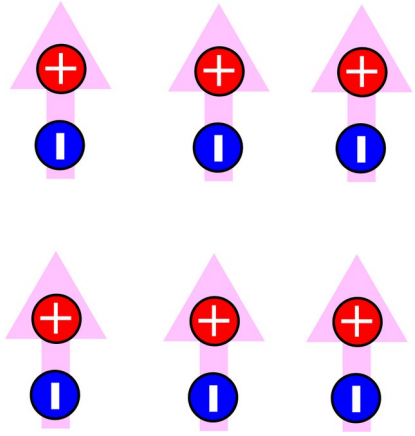
Classical magnets



$$\mathbf{M} \rightarrow -\mathbf{M}$$

Broken
rotational symmetry

Ferroelectrics



$$\mathbf{r} \rightarrow R\mathbf{r}$$

Broken
gauge symmetry

Superconductors



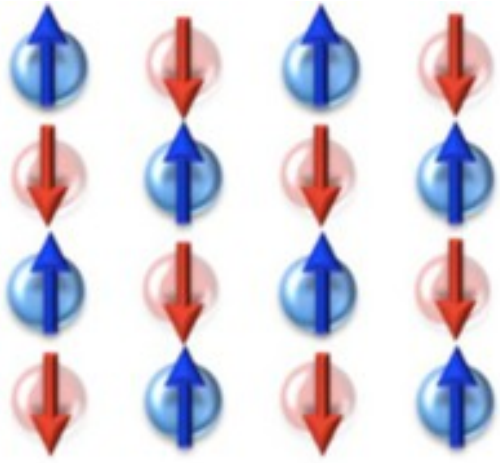
$$\langle c_\uparrow c_\downarrow \rangle \rightarrow e^{i\phi} \langle c_\uparrow c_\downarrow \rangle$$

How is symmetry related with quantum materials?

(Broken) symmetries allow to classify matter

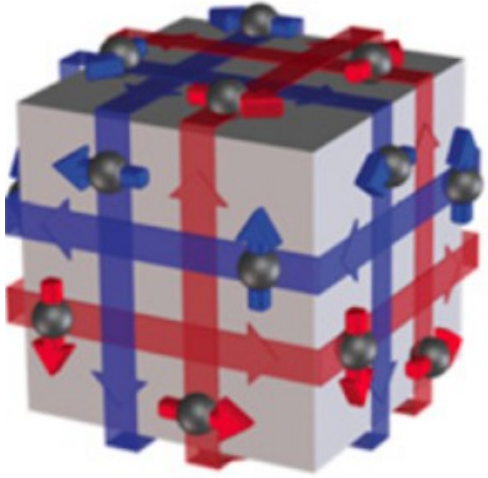
Symmetry classification

Ferromagnets
Superconductors
Ferroelectrics



Topological classification

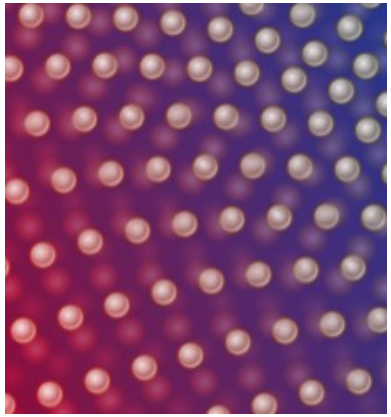
Trivial insulators
Topological insulators
Topologically ordered matter



Emergent excitations when symmetries get broken

Phonons

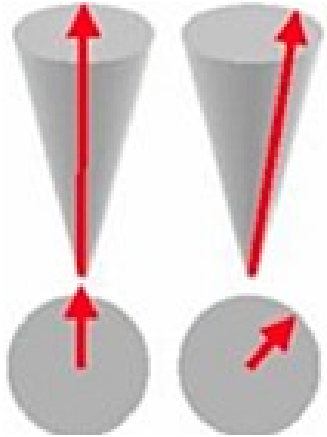
Crystals $\langle \vec{R} \rangle \neq 0$



Spin 0
Charge 0
Gapless

Magnons

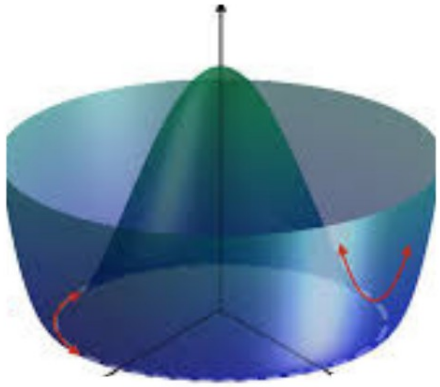
Magnets $\langle \vec{S} \rangle \neq 0$



Spin 1
Charge 0
Gapless/Gaped

Higgs mode

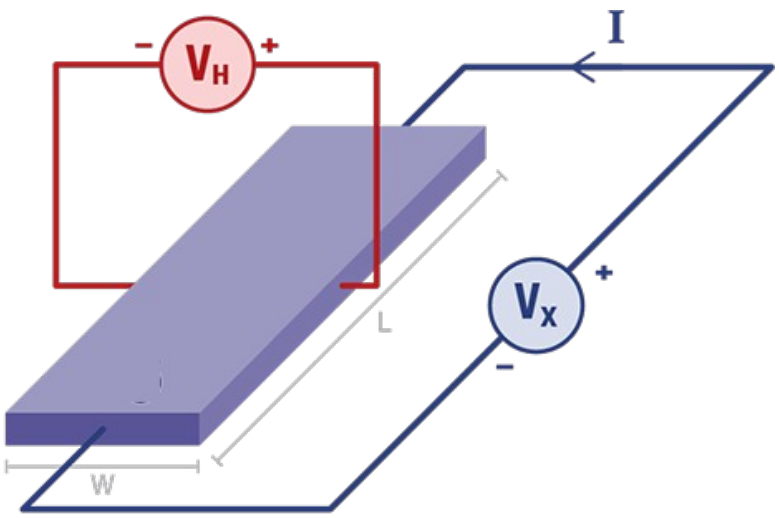
Superconductors $\langle c_\uparrow c_\downarrow \rangle \neq 0$



Spin 0
Charge 0
Gaped

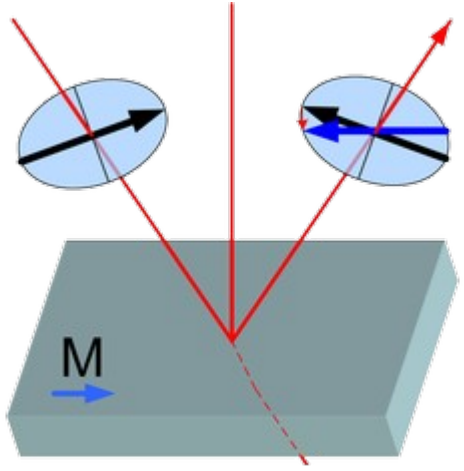
How to know if a material is magnetic?

Measure its Hall conductivity



Current generated perpendicular to a bias voltage

Measure its magneto-optical Kerr effect



Different reflection for left-handed and right-handed polarized light

What a material being “ferromagnetic” means?

An intuitive definition



“It sticks to your fridge”

A formal definition

$$\Theta|\Psi\rangle \neq |\Psi\rangle$$

Time reversal operator

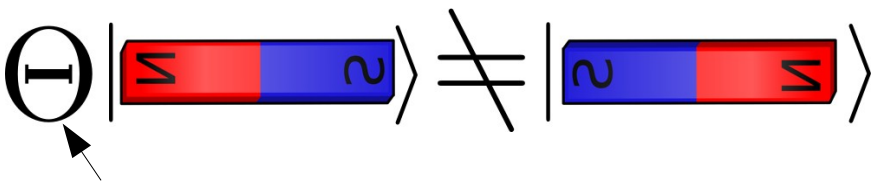
Wavefunction

It breaks time-reversal symmetry

A physical definition of a magnet

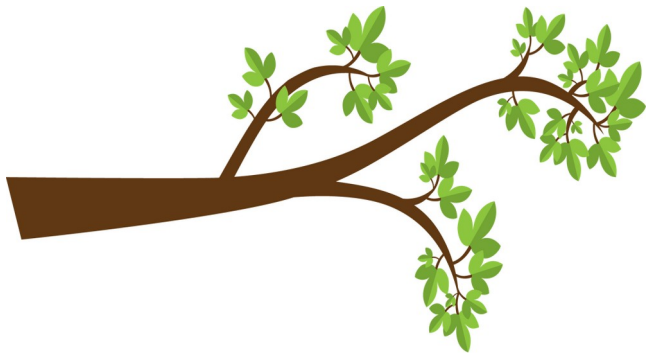
Magnet

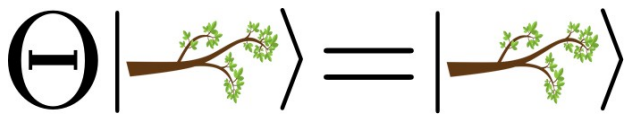




Time reversal operator

Not a magnet





Magnetic materials are not invariant under time reversal symmetry

A physical definition of a time-reversal broken magnet

Symmetry broken

$$|\Psi_T\rangle = |\uparrow\uparrow\rangle$$

$$\Theta|\uparrow\uparrow\rangle = |\downarrow\downarrow\rangle$$

Time reversal operator

Symmetry preserving

$$|\Psi_S\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right)$$

$$\Theta|\Psi_S\rangle = |\Psi_S\rangle$$

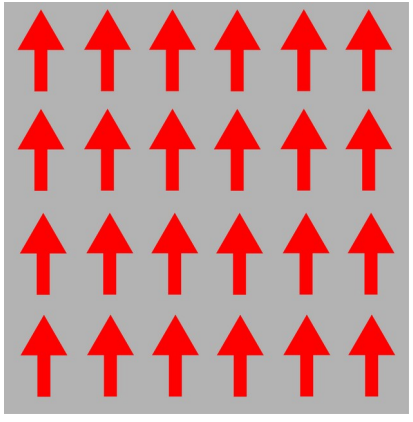
Many-body wavefunctions can be invariant or non-invariant under time-reversal

The role of electronic interactions

Electronic interactions are responsible for symmetry breaking

Broken time-reversal symmetry

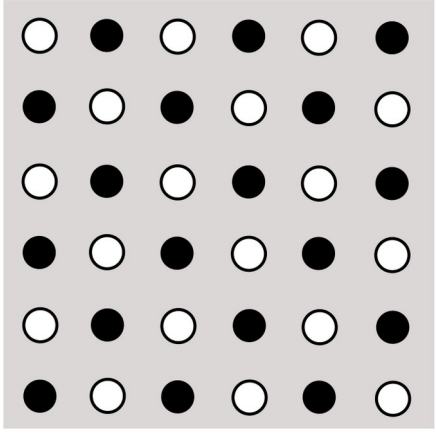
Classical magnets



$$\mathbf{M} \rightarrow -\mathbf{M}$$

Broken crystal symmetry

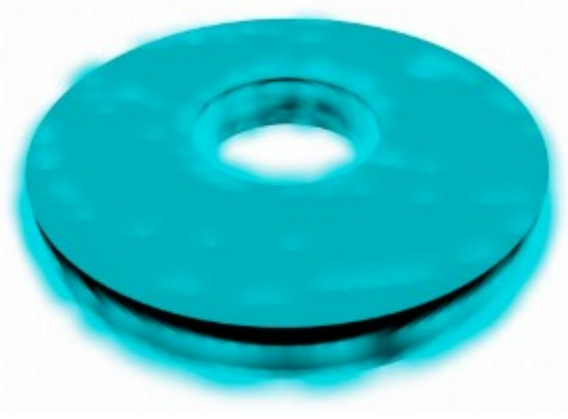
Charge density wave



$$\mathbf{r} \rightarrow \mathbf{r} + \mathbf{R}$$

Broken gauge symmetry

Superconductors



$$\langle c_\uparrow c_\downarrow \rangle \rightarrow e^{i\phi} \langle c_\uparrow c_\downarrow \rangle$$

Interactions and mean field

Free Hamiltonian ***Interactions***

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

What are these interactions coming from?

- Electrostatic (repulsive) interactions
- Mediated by other quasiparticles (phonons, magnons, plasmons,...)

The net effective interaction can be attractive or repulsive

Interactions promote a variety of correlated states

A spinless interacting Hamiltonian

Free Hamiltonian ***Interactions (nearest-neighbor repulsion)***

$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + V \sum_n \left(c_n^\dagger c_n - \frac{1}{2}\right)\left(c_{n+1}^\dagger c_{n+1} - \frac{1}{2}\right)$$

What is the ground state of this Hamiltonian?

The mean-field approximation, charge density waves

Mean field: *Approximate four fermions by two fermions times expectation values*

Four fermions (not exactly solvable) ***Two fermions (exactly solvable)***

$$V c_n^\dagger c_n c_{n+1}^\dagger c_{n+1} \approx V \langle c_n^\dagger c_n \rangle c_{n+1}^\dagger c_{n+1} + \ldots.$$

$$\sum_n V c_n^\dagger c_n c_{n+1}^\dagger c_{n+1} \approx \sum_n \mu_n c_n^\dagger c_n$$

Spatially dependent onsite energies

The selfconsistent problem in mean-field

The mean-field approximation is equivalent to finding a variational wavefunction of the form

$$|GS\rangle = \prod_\alpha \psi_\alpha^\dagger |\Omega\rangle$$

$$\psi_\alpha^\dagger = \sum_n \chi_{\alpha,n} c_n^\dagger$$

Such that it minimizes the expectation value of the energy

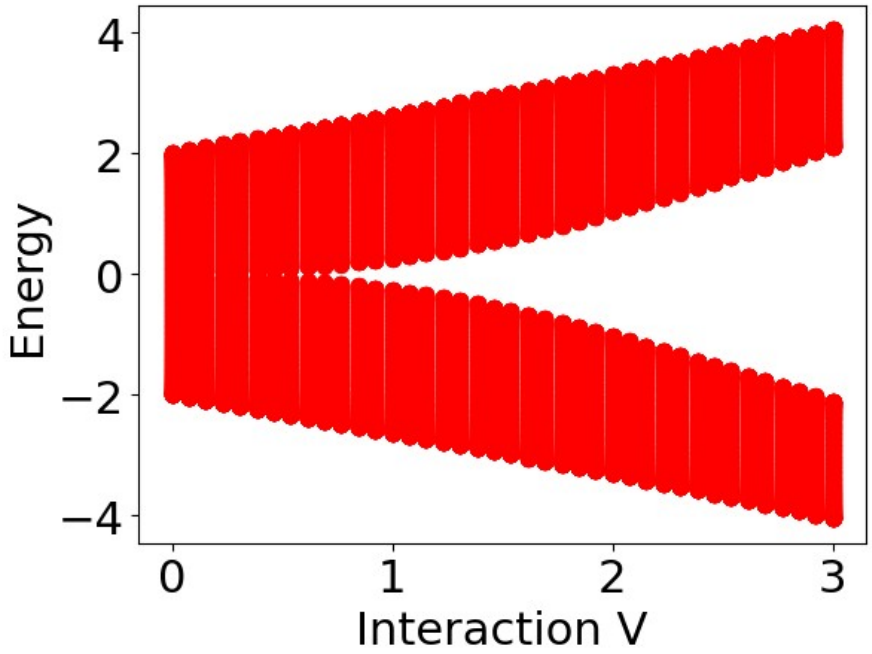
$$E = \langle GS|H|GS\rangle \qquad \frac{\delta E}{\delta \chi} = 0 \qquad \frac{\delta E}{\delta \langle c_i^\dagger c_j \rangle} = 0$$

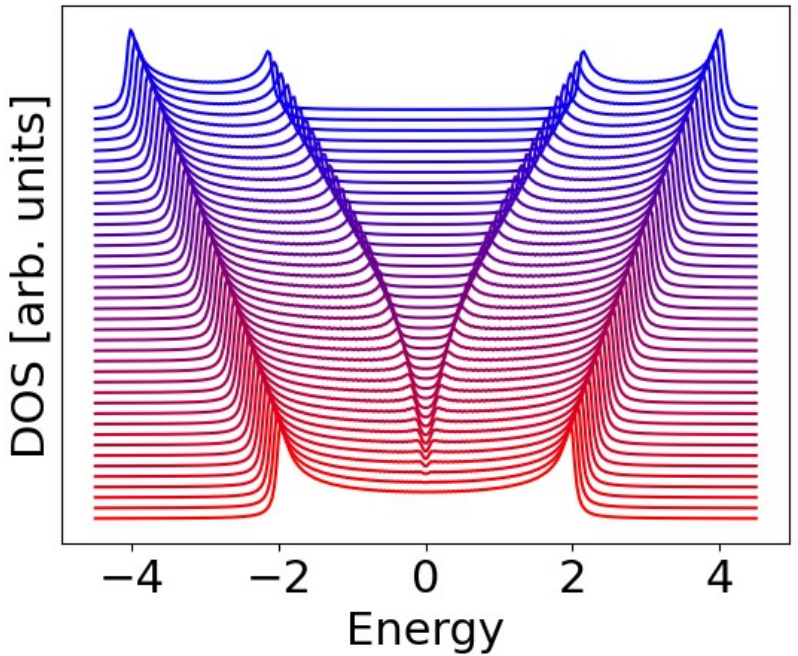
This assumption, leads to a selfconsistent problem for the mean-field Hamiltonian

$$H^{MF}_{(1)} \rightarrow \langle c_i^\dagger c_j \rangle_{(1)} \rightarrow H^{MF}_{(2)} \rightarrow \langle c_i^\dagger c_j \rangle_{(2)} \rightarrow H^{MF}_{(3)} \rightarrow \ldots$$

The impact of non-local interactions in the electronic spectra

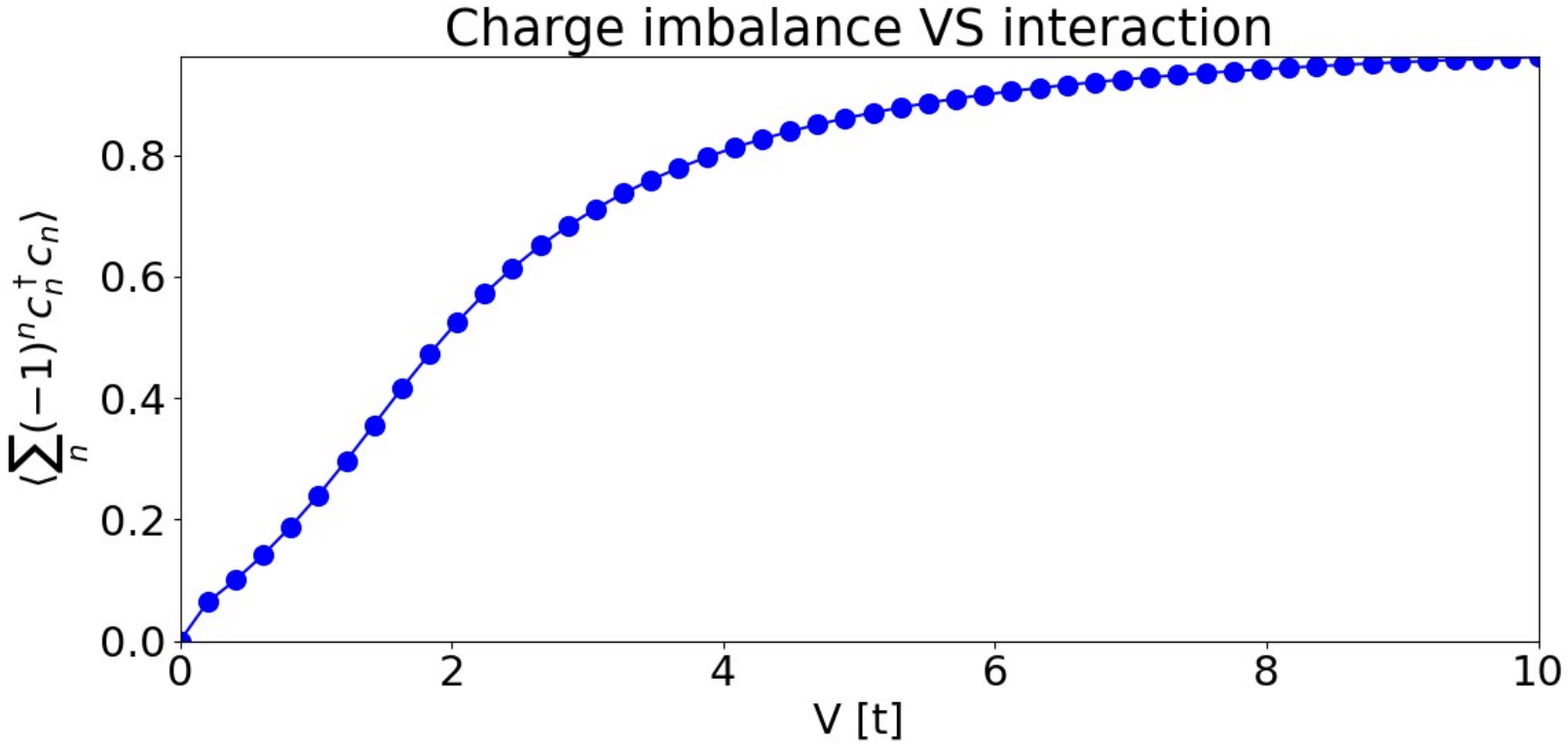
$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + V \sum_n \left(c_n^\dagger c_n - \frac{1}{2}\right)\left(c_{n+1}^\dagger c_{n+1} - \frac{1}{2}\right)$$





Interaction induced charge density wave

$$H = \sum_n c_n^\dagger c_{n+1} + h.c. + V \sum_n \left(c_n^\dagger c_n - \frac{1}{2}\right)\left(c_{n+1}^\dagger c_{n+1} - \frac{1}{2}\right)$$



A spinful interacting Hamiltonian

Free Hamiltonian ***Interactions (Hubbard term)***

$$H = \sum_{ij} t_{ij} [c^\dagger_{i\uparrow} c_{j\uparrow} + c^\dagger_{i\downarrow} c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

From now on lets consider we have a spin degree of freedom $\uparrow, \downarrow$

What is the ground state of this Hamiltonian?

$U < 0$ Superconductivity

$U > 0$ Magnetism

The mean-field approximation, magnetism

Mean field: *Approximate four fermions by two fermions times expectation values*

Four fermions (not exactly solvable) ***Two fermions (exactly solvable)***

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx U \langle c^\dagger_{i\uparrow} c_{i\uparrow} \rangle c^\dagger_{i\downarrow} c_{i\downarrow} + ... + h.c.$$

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx M \sigma^z_{ss'} c^\dagger_{i,s} c_{i,s'} + h.c.$$

For $U > 0$
i.e. repulsive interactions

Magnetic order
$$M \sim \langle c^\dagger_{i\uparrow} c_{i\uparrow} \rangle - \langle c^\dagger_{i\downarrow} c_{i\downarrow} \rangle$$

Time-reversal symmetry and magnetism

Time-reversal symmetry switches the arrow of time in the Hamiltonian

It reverses the angular momenta, and in particular the spin $\vec{S} \rightarrow -\vec{S}$

Time-reversal symmetry switches up and down spins

$$c^\dagger_{n,\uparrow} c_{n,\uparrow} \rightarrow c^\dagger_{n,\downarrow} c_{n,\downarrow}$$
$$c^\dagger_{n,\downarrow} c_{n,\downarrow} \rightarrow c^\dagger_{n,\uparrow} c_{n,\uparrow}$$

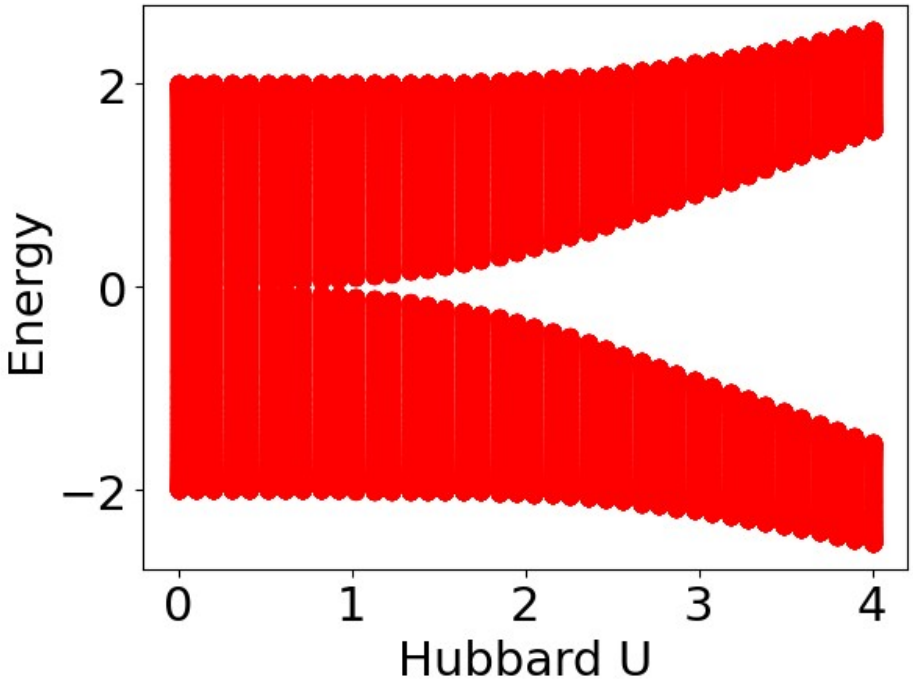
How does the magnet transform under time reversal symmetry?

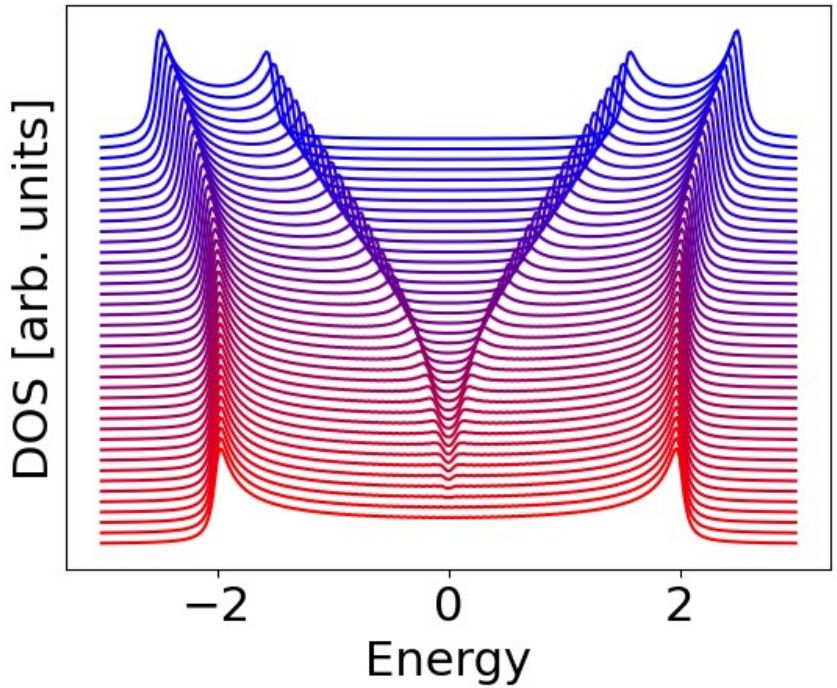
$$M_z \sim \langle c^\dagger_{n,\uparrow} c_{n,\uparrow} \rangle - \langle c^\dagger_{n,\downarrow} c_{n,\downarrow} \rangle \qquad M_z \rightarrow -M_z$$

A magnet breaks time-reversal symmetry

Interaction-induced gap opening

$$H = t\sum_{s,n} c^{\dagger}_{n,s}c_{n+1,s} + h.c. + U\sum_{n}\left(c^{\dagger}_{n,\uparrow}c_{n,\uparrow} - \frac{1}{2}\right)\left(c^{\dagger}_{n,\downarrow}c_{n,\downarrow} - \frac{1}{2}\right)$$





$U = 4t$

$U = 0$

The mean-field approximation, superconductivity

Mean field: *Approximate four fermions by two fermions times expectation values*

Four fermions (not exactly solvable) ***Two fermions (exactly solvable)***

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx U \langle c^\dagger_{i\uparrow} c^\dagger_{i\downarrow} \rangle c_{i\uparrow} c_{i\downarrow} + h.c.$$

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx \Delta c_{i\uparrow} c_{i\downarrow} + h.c.$$

For $U < 0$

i.e. attractive interactions

$\Delta \sim \langle c^\dagger_{i\uparrow} c^\dagger_{i\downarrow} \rangle$ is the superconducting order

Gauge symmetry and superconductivity

What we know from quantum mechanics

"The phase of a wavefunction (field operator) does not have physical meaning"

This is what we know as gauge symmetry

$$c_n \to e^{i\phi} c_n$$

$$c_n^\dagger \to e^{-i\phi} c_n^\dagger$$

How does the superconducting pairing transform under a gauge transformation?

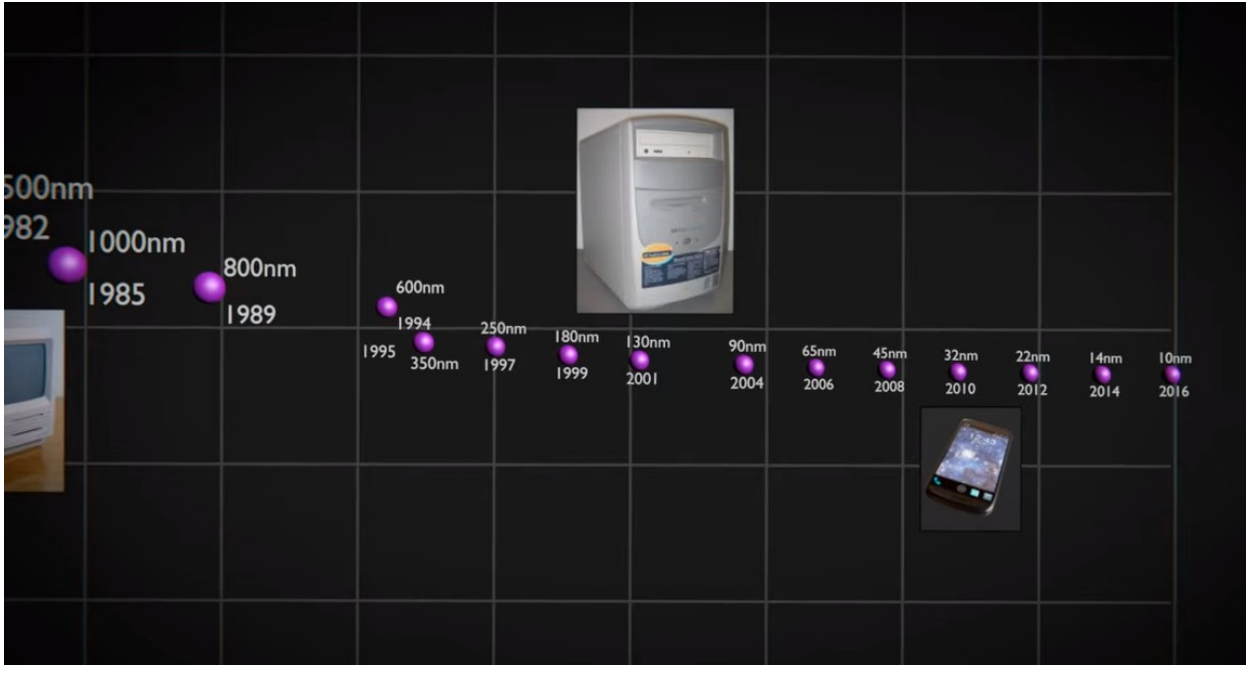
$$\Delta \sim \langle c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger \rangle$$

$$\Delta \to e^{-2i\phi} \Delta$$

A superconductor breaks gauge symmetry

An overview of the impact of quantum materials in society

https://www.youtube.com/watch?v=3De1rLxvzyU



Take home

- Quantum materials realize a wide range of exotic phenomena
- Second quantization is a versatile language to understand quantum phenomena
- Symmetry breaking leads to new quantum states